EMMANUEL SAMPSON

• ENGINEERING STUDENT • PHYSICS RESEARCHER

CONTACT

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EDUCATION

William & Mary
BS Engineering Physics and Applied Design
BA Philosophy with Honors
Expected Completion May 2026

UCLA School of Theater, Film & Television
Creative Production Program
Completed July 2024

Achievements

Winner of William & Mary 2024 and 2025 Hackathon

William and Mary Spirit Scholarship 2024
Selected as a student who exemplifies the William & Mary spirit: leadership, community and international service.

William and Mary Catron Scholarship
Awarded to students who represent great potential and ambition within the arts.

William and Mary Bosanko Scholarship

Sharpe Community Scholar

Marine Officer Candidate School Attendant

KEY SKILLS

Python, HTML, SPSS

Jira Service Manager

Microsoft Office, Microsoft Excel

Fiber Optics & RF Engineering

Autodesk Suite Fusion 360, Eagle (CAD)

Data Analysis & Signal Processing

Technical Writing

Computer Vision, Artificial Intelligence, and Machine Learning, Neural Networks

Conversational Japanese

High Performance Computing (HPC)

Multisim

KiCad

Certifications

The Boehly Center for Excellence in Finance Introduction To Python Training

From Excel to Tableau

Everybody's Introduction to Tableau (2022)

Tableau for Data Scientists

Corporate Finance: Environmental, Social, and Governance (ESG)

EXPERIENCE

ENGINEER INTERN — MAY 2025—PRESENT

WEINERT Fiber Optics Inc • Williamsubrg, VA

Designed and implemented AI-powered computer vision systems in Python and C++ for automated image analysis and defect detection on fiber-optic end-faces.
Developed, trained, and deployed Convolutional Neural Networks (CNNs) using High-Performance Computing (HPC) clusters to accurately locate and identify microscopic surface defects.
Engineered a custom microscope and proprietary software program that automated the full inspection process, enhancing data transmission efficiency for critical medical applications, including instrumentation for kidney stone removal and cancer treatment.

PHYSICS RESEARCH ASSISTANT — OCT 2023—PRESENT

Applied Research Center Core Labs • Williamsburg, VA

Managed a $4,000 budget for the procurement of new electronics and essential laboratory equipment, overseeing the entire acquisition lifecycle from needs analysis to purchasing.
Supported key research initiatives by operating classical and electron microscopes (SEM/TEM), performing sample analysis, and organizing complex experimental data sets, while concurrently conducting independent research in remote sensing and reflective optics.
Trained and mentored graduate students and industry professionals on the operation, theory, and application of advanced material characterization instrumentation.

STUDENT MAKERSPACE ENGINEER — OCT 2022—NOV 2023

W&M Makerspace • Williamsburg, VA

Managed and resolved over 1,000 service and technical support requests using the Atlassian Jira ticketing system for advanced digital fabrication equipment, including laser cutters, 3D printers, and CNC mills.

Trained and certified over 200 students on safety protocols and the proficient operation of advanced manufacturing machinery, providing expert technical consultation for research and academic projects.

Designed and led a multi-level robotics workshop series, developing a tiered curriculum and instructional materials to effectively train participants from beginner to advanced proficiency.

CAMPUS INVOLVEMENT AND EXTRACURRICULARS

Robotics Club President
Led weekly robotics workshops for an average of 20 students, developing a tiered curriculum to engage participants from beginner to advanced skill levels.
Launched and directed the inaugural William & Mary STEM Symposium, securing 15 industry-leading scientists and engineers as speakers and drawing an audience of over 150 students.
Designed a semester-long project curriculum covering foundational and advanced topics, including analog/digital electronics, radio frequency (RF) devices, and remote control systems.

Data Science Society President
Manage and facilitate weekly technical workshops on key data science tools and methodologies, including sessions on Power BI, Kaggle, and ArcGIS.
Expanded the society's curriculum by designing and launching advanced workshops focused on Computer Vision, Machine Learning (ML), and Neural Networks.

Technical Executives Association President
Preside over a leadership council of 8 technical student organizations (including societies for Robotics, Rocketry, Data Science, Computer Science, and AI), fostering inter-organizational collaboration and aligning strategic goals.

Projects

Bird Sensor Hackathon 2024 – First Place Winner
Developed and pitched a novel IoT device to prevent avian-window collisions by detecting and analyzing the unique electromagnetic signatures of bird flight patterns. Engineered a predictive model using proximity and sensor data to anticipate potential impacts in real-time, triggering a collision avoidance or deterrent system.

WaveFinderAI Hackathon 2025 – 2nd Place Winner
Designed and simulated a machine learning tool to detect and classify microplastic contaminants in fluid systems by analyzing their unique reflective frequencies.
The model provided real-time data analysis of fluid velocity, contaminant size and shape, and concentration levels, with direct applications in predictive maintenance for industrial hydraulics.

Senior Project 2025 Automated Desktop SMD
Designing .Portable Desktop Size Surface Mounting Device(SMD) to contribute to development of microwave circuits, 3D PCB-like boards that move nanomaterials with microwave frequencies.